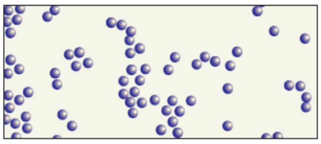
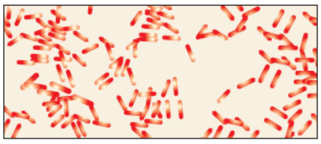
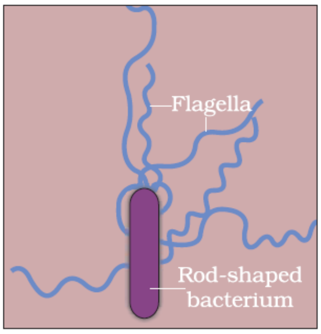
Introduction

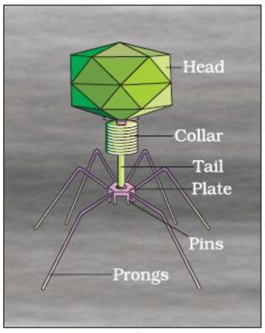
Besides macroscopic plants and animals, microbes are the major components of biological systems on this earth. Microbes are present everywhere – in soil, water, air, inside our bodies and that of other animals and plants. They are present even at sites where no other life-form could possibly exist–sites such as deep inside the geysers (thermal vents) where the temperature may be as high as 1000C, deep in the soil, under the layers of snow several metres thick, and in highly acidic environments. Microbes are diverse–protozoa, bacteria, fungi and microscopic plants viruses, viroids and also prions that are proteinaceous infectious agents. Some of the microbes are shown below. Microbes like bacteria and many fungi can be grown on nutritive media to form colonies that can be seen with the naked eyes. Such cultures are useful in studies on micro-organisms.


Spherical shaped bacteria, magnified 1500X

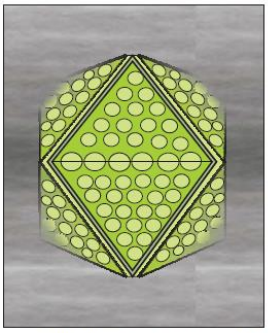

Rod-shaped bacteria, amgnified 1500X

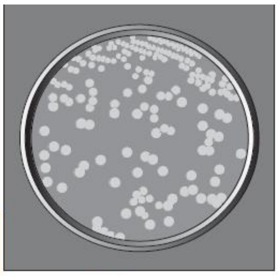


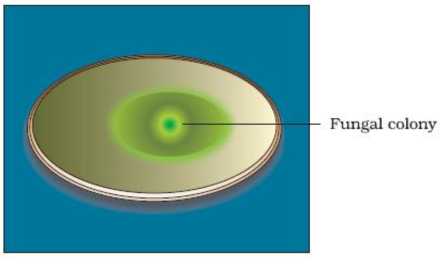
A rod-shaped bacterum showing flagella, magnified 50,000X



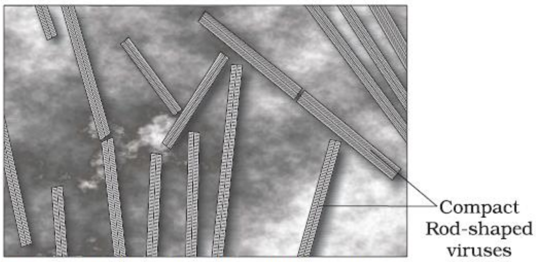
A bacteriophage


Adenovirus which causes repiratory infection


Colonies of bacteria growing in a petri dish



Fungal colony growing in a petri dish



Rod-shaped Tobacco Mosaic Virus (TMV), amgnified about 1,00,000 - 1,50,000X

Microbes in Household products:

1. Lactic acid bacteria (LAB)

- A common example is the production of curd from milk. Micro-organisms such as *Lactobacillus* and others commonly called **Lactic Acid Bacteria (LAB)** grow in milk and convert it to curd. During growth, the LAB produces acids that coagulate and partially digest the milk proteins. Small amount of curd is added to the milk for curdling acts as an inoculum containing thousands of LABS, which further multiply. It also improves its nutritional quality by increasing **vitamin B12**. In our stomach too, the LAB play very beneficial role in checking disease causing microbes.

2. Fermentation

- Dosa and idli dough is fermented by bacteria, which produces CO2 gas and give it a puffed-up appearance.
- The dough, which is used for making bread, is fermented by using **baker's yeast** (*Saccharomyces cerevisiae*).
- **"Toddy",** a traditional drink of some parts of southern India is made by fermenting sap from palms.

3. Cheese making

- Microbes are also used to ferment fish, soya bean and bamboo-shoots to make foods. Cheese, is one of the oldest food items in which microbes were used. The large holes in **"Swiss cheese"** are due to production of a large amount of CO2 by a bacterium named *Propionibacterium sharmanii*. The **"Roquefort cheese"** is ripened by growing a specific fungus on them for a particular flavour.

Microbes in Industrial Products – Industrial applications

- Production on an industrial scale requires growing microbes in very large vessels called **Fermentors**.
- On industrial scale. Fermented beverages, antibiotics, enzymes and other bioactive molecules are prepared using microbes.

Fermentation Plant

Fermentor

1. Fermented Beverages:

- The yeast *Saccharomyces cerevisiae* used for bread making and commonly called **brewer's yeast**, is used for fermenting malted cereals and fruit juices to produce **ethanol**. Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth.

2. Antibiotics (Gr. Anti = against; Bios = life) (in the context of disease causing organisms)

- Antibiotics are chemical substances, which are produced by some microbes and can kill or retard the growth of other disease causing microbes.
- **Penicillin** was the first antibiotic to be discovered and it was a chance discovery. **Alexander Fleming** while working on *Staphylococci* bacteria, once observed a mould growing in one of his unwashed culture plates around which *Staphylococci* could not grow. He found out that it was due to a chemical produced by the mould and he named it Penicillin after the mould *Penicillium notatum (Penicillium chrysogenum)*. Later, **Ernest Chain** and **Howard Florey** made its full potential effective antibiotic.

NOTE - Penicillin was extensively used to treat American soldiers wounded in World War II. Fleming, Chain and Florey were awarded the Nobel Prize in 1945, for this discovery. Antibiotics have greatly improved our capacity to treat deadly diseases such as plague, whooping cough (kali khasi), diphtheria (gal ghotu) and leprosy (kusht rog), which used to kill millions all over the globe.

3. Chemicals, Enzymes and other Bioactive Molecules:

- Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes.
- Examples of acid producers are -
  - *Aspergillus niger* (fungus) – **Citric acid**
  - *Acetobacter aceti* (bacterium) – **Acetic acid**
  - *Clostridium butylicum* (bacterium) – **Butyric acid**
  - *Lactobacillus* (bacterium) – **Lactic acid**
  - *Saccharomyces cerevisiae* (Yeast - unicellular fungus) – is used for commercial production of Ethanol.

Microbes used for the production of Enzymes:

- **Lipase** – used in laundry detergents and helps in removing oily stains from the laundry.
- **Pectinase** and **protease** – used in bottled juices to clarify the juices
- **Streptokinase** (Streptococcus bacterium) – used as **clot buster** (to remove clots) for removing clots from the blood vessels of patients who have undergone myocardial infraction leading to heart attack.

Microbes used to produce bioactive molecules:

- **Cyclosporin A** (*Trichoderma polysporum* **-** fungi) – used as immunosuppressive agent (for organ transplant patients).
- **Statins** (*Monascus purpureus* **-** yeast) – used as blood cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Microbes in Sewage Treatment:

- Treatment of waste water is done by **heterotrophic microbes** naturally present in the sewage.
- Before disposal, hence, sewage is treated in **sewage treatment plants** (STPs).
- This treatment is carried out in two stages;

1. **Primary treatment / Physical treatment:** It involves physical removal of particles – large and small - from the sewage through filtration and sedimentation.
    - Initially, **sequential filtration** is used to remove floating debris
    - Then, **girt** (soil and small pebbles) are removed by **sedimentation**.
    - All solids that settle form the **primary sludge**, and the supernatant forms the **effluent**. The effluent from the primary settling tank is taken for secondary treatment.

2. **Secondary treatment / Biological treatment:**
    - The primary effluent is passed into large aeration tanks, this allows vigorous growth of aerobic microbes into **flocs** (masses of bacteria associated with fungal filaments to form mesh like structure).
    - While growing, these microbes consume the major part of the organic matter in the effluent. This significantly reduces the **BOD (biochemical oxygen demand)** of the effluent.
    - **BOD refers to the amount of the oxygen that would be consumed if all the organic matter in one liter of water oxidised by bacteria** (BOD is the measure of the organic matter present in the water). The greater the BOD of waste water, more is its polluting potential.
    - Once the BOD of sewage water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called **Activated sludge**.
    - A small part of this activated sludge is pumped back into the aeration tank to serve as the inoculum.
    - The remaining major part of the sludge is pumped into large tanks called **anaerobic sludge digesters.** Here, other kinds of bacteria which grow anaerobically, digest the bacteria and the fungi in the sludge.
    - During this digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form **biogas**.
    - The effluent from the secondary treatment plant is generally released into natural water bodies like rivers and streams.

Secondary treatment

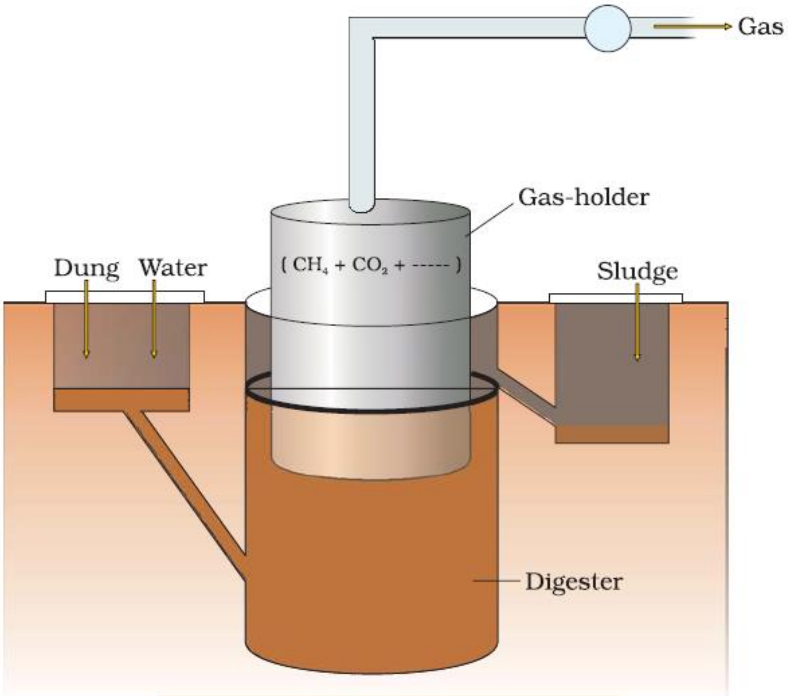
NOTE – The Ministry of Environment and Forests has initiated **Ganga Action Plan** and **Yamuna Action Plan** to save these major rivers of our country from pollution. Under these plans, it is proposed to build a large number of sewage treatment plants so that only treated sewage may be discharged in the rivers.

Microbes in Production of Biogas

- Biogas is mixture of gases (containing predominantly methane) produced by the microbial activity and which may be used as fuel.
- The type of gas produced depends upon the substrate they grow in and the type of microbe.
- Certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO2 and H2. These bacteria are collectively called **Methanogens** (Methanobacterium).
- These bacteria are also present in the rumen of cattle. A lot of cellulosic material present in the food of cattle is also present in the rumen. In rumen, these bacteria help in the breakdown of cellulose and play an important role in the nutrition of cattle.
- Thus, the excreta (dung) of cattle, commonly called **Gobar**, is rich in these bacteria.
- Dung can be used for generation of biogas commonly called **gobar gas**.

Biogas Plant:

- The technology of biogas production was developed in India mainly due to the efforts of **Indian Agricultural Research Institute (IARI)** and **Khadi and Village Industries Commission (KVIC).**
- The biogas plant consists of a 10 – 15 feet deep **concrete tank** in which bio-wastes are collected and slurry of dung is fed.
- A **floating cover** is placed over the slurry, which keeps on rising as the gas is produced in the tank due to the microbial activity.
- The biogas plant has an **outlet**, which is connected to a **pipe** to supply biogas to nearby houses.
- The spent slurry is removed through another outlet and may be used as fertilizer.
- The biogas thus produced is used for cooking and lighting.
- Biogas plant is usually set up in rural areas since cow dung is available in abundance there.



A typical biogas plant

Microbes as Bio control Agents:

- Chemical insecticides and pesticides are harmful as
  - They kill the useful and harmful life forms indiscriminately
  - They are toxic to human beings as well as in the long run
  - If all insects of a particular species are killed, then the natural predator-prey relationship and food chains will get distorted
- Hence, biological means to eradicate pests can be used. This requires knowledge of the life forms (predator and prey) that inhabit a particular area, their life cycles and patterns of feeding and preferred habitats.
- Example - Ladybirds and dragonflies are used to get rid of aphids and mosquitoes
- Microbes can also act in the same manner. *Bacillus thuringiensis* (*Bt*) is used to control butterfly caterpillars.
- This bacterium is available in scahets as dried spores, which are sprayed on the crops. The spores get into the gut of the larvae and kill it while the other insects remain unperturbed.
- By methods of genetic engineering, the genes of *B. thuringiensis* responsible for killing the larvae have been incorporated into the plants
- Cotton plant with *Bt* gene incorporated is called *Bt*-cotton
- The fungus **Trichoderma** living in roots of plants acts as a bio control agent against several plant pathogens.
- *Baculoviruses*, particularly genus *Nucleopolyhedrovirus*, are also used as a narrow spectrum insecticidal agents.
- Bio control agents are particularly useful when insects are required to be conserved under **IPM (integrated pest management programmes)**.

Microbes as bio-fertilizers

- Chemical fertilizers contribute to the pollution.
- Bio-fertilizers are organisms that enrich the nutrient quality of the soil.
- Many bacteria, fungi and cyanobacteria act as **biofertilizers**.
- They act as bio-fertilizers by living in symbiotic association with root nodules of leguminous plants such as *Rhizobium*.
- These bacteria fix atmospheric nitrogen and enrich the nitrogen content of soil.
- Fungi also form symbiotic associations with plants **(Mycorrhiza)** such as many members of the genus *Glomus* form mycorhiza. They absorb phosphorus and pass it to plants.
- Cyanobacteria such as *Nostoc*, *Anabaena* etc. also fix atmospheric nitrogen and act as bio-fertilizers especially in paddy fields.

SUMMARY

Microbes are a very important component of life on earth. Not all microbes are pathogenic. Many microbes are very useful to human beings. We use microbes and microbially derived products almost every day. Bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd. The dough, which is used to make bread, is fermented by yeast called *Saccharomyces cerevisiae*. Certain dishes such as *idli* and *dosa*, are made from dough fermented by microbes. Bacteria and fungi are used to impart particular texture, taste and flavor to cheese. Microbes are used to produce industrial products like lactic acid, acetic acid and alcohol, which are used in a variety of processes in the industry. Antibiotics like penicillins produced by useful microbes are used to kill disease-causing harmful microbes. Antibiotics have played a major role in controlling infectious diseases like diphtheria, whooping cough and pneumonia. For more than a hundred years, microbes are being used to treat sewage (waste water) by the process of activated sludge formation and this helps in recycling of water in nature. Methanogens produce methane (biogas) while degrading plant waste. Biogas produced by microbes is used as a source of energy in rural areas. Microbes can also be used to kill harmful pests, a process called as biocontrol. The biocontrol measures help us to avoid heavy use of toxic pesticides for controlling pests. There is a head need these days to push for use of biofertilisers in place of chemical fertilisers. It is clear from the diverse uses human beings have put microbes to that they play an important role in the welfare of human society.